



Iranian Mathematical Society Competition

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by mojyla222

– Day 1

P1 Prove that there does not exist a sequence $x_1, x_2, \dots$ of real numbers such that:

$$\lim_{n \rightarrow \infty} (x_{n+1} + x_{n+2} + x_{n+3}) = 7$$

and

$$\lim_{n \rightarrow \infty} (x_{n+1} + x_{n+2} + \dots + x_{n+7}) = 3.$$

P2 Let R be a ring. If for every $x, y \in R$ we have $(x+y)^2 = x^2 + y^2$, then prove that for every $x, y \in R$:

$$(xy)^2 = x^2 y^2.$$

P3 Let (Ω, P) be a finite probability space, and let $E_1, E_2, \dots, E_n$ be events in Ω . Suppose that every element $x \in \Omega$ has the property that if $x \in E_i$, then x belongs to at most a_i of the events $E_1, E_2, \dots, E_n$. Prove that:

$$\sum_{i=1}^n \frac{P(E_i)}{a_i} \leq 1.$$

P4 Let G be a finite group such that for any two subgroups H and K of G , the order of $H \cap K$ is equal to the greatest common divisor of $|H|$ and $|K|$. Prove that G is a cyclic group.

P5 A complete metric space (X, d) has the property that for every sequence $\epsilon_1, \epsilon_2, \dots$ of positive real numbers, there exists a sequence $x_1, x_2, \dots$ of points in X such that:

$$X = \bigcup_{i=1}^{\infty} B_{\epsilon_i}(x_i),$$

where $B_{\epsilon}(x)$ denotes the open ball of radius ϵ centered at x . Prove that X is a countable set.

- P6** Ayla has 100 distinct real numbers $x_1, x_2, \dots, x_{100}$. Bardia's goal is to find the index k such that:

$$x_k = \min\{x_1, x_2, \dots, x_{100}\}.$$

To achieve this, Bardia selects two indices i and j from the set $\{1, 2, \dots, 100\}$ in each step and asks Ayla, "Which of the numbers x_i or x_j is smaller?" After Ayla answers Bardia's question, Bardia proceeds to the next question, and the game continues in this manner. Ayla is allowed to give at most one incorrect answer. Determine the minimum number of questions Bardia needs to ask to guarantee achieving his goal. Prove your claim.

– Day 2

- P7** Let n be a natural number, and let $\{a_1, \dots, a_n\}$ and $\{b_1, \dots, b_n\}$ be two sets of real numbers. If for every real number t , we have:

$$\sum_{i=1}^n |a_i - t| \leq \sum_{i=1}^n |b_i - t|,$$

prove that:

$$\sum_{i=1}^n a_i = \sum_{i=1}^n b_i.$$

- P8** Let A be a real square matrix. If $A - A^2$ is nilpotent, prove that the characteristic polynomials of A and A^2 are equal.

- P9** Let (X, d) be a compact metric space, and let $T : X \rightarrow X$ be a continuous function with the property that for any two distinct points $x, y \in X$, there exists a natural number n such that:

$$d(T^n(x), T^n(y)) < d(x, y).$$

(Here, T^n denotes the n -th iterate of T .) Prove that there exists a point $x_0 \in X$ such that $T(x_0) = x_0$.

- P10** Let $n \geq 2$ be a natural number, and let A be the set of all vectors of length n with components in $\{0, 1, \star\}$. For any two vectors $(x_1, \dots, x_n)$ and $(y_1, \dots, y_n)$ in A , define the distance between them as the number of indices i (from 1 to n) for which $\{x_i, y_i\} = \{0, 1\}$. Prove that the maximum number of elements in A such that the pairwise distance between any two is at least 1 and at most $n - 1$ is equal to $3 \times 2^{n-2}$.

- P11** Let n be a natural number, and let $z_1, z_2, \dots, z_n$ be distinct complex numbers with absolute value 1. Define:

$$P(z) = (z - z_1)(z - z_2) \cdots (z - z_n).$$

For every real number $r > 1$, prove that there exists $z_0 \in \mathbb{C}$ such that $|z_0| = r$ and $|P(z_0)| = r^n$.

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- P12** Let R be a commutative ring admitting identity element. An ideal I of R is called *large* if I has a non-trivial intersection with every non-zero ideal of R . Suppose that every large ideal in R contains an element that is not a zero-divisor. Prove that the sum of all minimal ideals in R is generated by a single idempotent element.
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